Tóm tắt tacn

BOOK 1

Chương 2-Introduction to Computing Systems

Phân loại máy tính:

Type	Description	Used For
Desktop	A personal computer designed to be used at a single location.	
PC Handheld	Hybrid console	PDAs
Microcomputer	A system for a PC; includes desktops, laptops, etc.	Nó cũng là máy tính cá nhân vs laptop
Minicomputer	Medium-sized computer; stronger than micro, weaker than mainframe	-Managing business operations -Data collection and analysis -Industrial control -Multi-user systems
Workstation	High-performance desktop for specialized tasks	-Graphic design & video editing -3D modeling and animation -Software development -Scientific research
Server	Provides data, services, or resources over a network	-Hosting websites & databases -File storage/sharing -Email/cloud services – -User access control
Mainframe	Very large, powerful computer for large organizations	-Transaction processing (e.g., banking) -Big data storage -Enterprise apps -Airline/insurance systems

Chương 3-Inside the Computer

Từ viết tắt về các linh kiện và nhận diện

Từ	Full từ	Từ nhận diện
CPU	**Central Processing Unit**	
RAM	**Random Access Memory**	**Temporary storage**
ROM	**Read-Only Memory**	**Permanent instructions**
ALU	**Arithmetic Logic Unit**	**Calculations and logic**
CU	**Control Unit**	**Coordination and control**
USB	**Universal Serial Bus**	**Peripheral connection**
GHz	**Gigahertz**	**CPU speed**
BIOS	**Basic Input/Output System**	**Startup and hardware check**
HDD	**Hard Disk Drive**	**Mechanical storage**
SSD	**Solid State Drive**	**Fast storage**
PSU	**Power Supply Unit**	**Power provider**
GPU	**Graphics Processing Unit**	
MB	**Motherboard**	**Component hub**

Vocabulary

Heat sink a part used to cool the processor

Expansion card a part used to increase a computer’s functions

Processsor a part that completes tasks for the computer

Hard drive a part used to store large amounts of data

Motherboard a slotted board with circuits and ports

Chương 4-Computing Devices

Từ viết tắt về các linh kiện và nhận diện

Chương này thêm mấy cái thiết bị input output nữa mà nó cơ bản quá k ném zô đây nhe

Wi-Fi Wireless Fidelity	Wireless network connection
LCD Liquid Crystal Display	Thin display screen
LED Light Emitting Diode	Energy-saving screen
IoT Internet of Things	Smart connected devices
GPS Global Positioning System	Location tracking
OS Operating System	Hardware-software interface
AI Artificial Intelligence	Smart automation / virtual assistant

Các loại bộ nhớ:

Loại bộ nhớ	Vai trò	Các thiết bị tiêu biểu
Primary Storage	**Temporarily stores data being processed** -Very fast access speed	-RAM (Random Access Memory) -ROM (Read-Only Memory) -Cache Memory -CPU Registers
Secondary Storage	**Stores data permanently** -Retains data after power off	-HDD (Hard Disk Drive) -SSD (Solid State Drive) -- SSHD (Hybrid Drive)
Removable Storage	**External storage devices** -Used for backup or data transfer	-USB flash drive -Memory card (SD card, microSD) -External HDD/SSD -- CD/DVD/Blu-ray
Tertiary Storage	**Used for large-scale, long-term storage** -Infrequently accessed	-Magnetic tape (Tape Drive) -Optical jukebox -- Archival cloud storage (e.g. Amazon Glacier)

Vocabulary

Keyboard a device that enters information into a computer

Mouse a device is used to control a pointer or cursor

Scanner a device that reads images and sends them to a computer

Mircophone a device that gathers sounds

Monitor a device that displays images

Chương 5-Networking

A computer on Internet is uniquely identified by I**P address**

ISP is a company that provides users with access to the internet.

Viết tắt và nhận diện các từ của chương

Từ	Full từ	Từ nhận diện
LAN	Local Area Network	Local network (home, office)
WAN	Wide Area Network	Large-scale network
MAN	Metropolitan Area Network	City-scale network
IP	Internet Protocol	Device identification, data routing
URL	Uniform Resource Locator	Website address
HTTP	HyperText Transfer Protocol	Web communication
HTTPS	HTTP Secure	Secure web communication
ISP	Internet Service Provider	Internet provider
Wi-Fi	Wireless Fidelity	Wireless internet
DSL	Digital Subscriber Line	Internet over phone line
IP address	Internet Protocol Address	Network device identifier
DNS	Domain Name System	Domain to IP translation
MAC	Media Access Control	Hardware address
VPN	Virtual Private Network	Secure online connection
TCP/IP	Transmission Control Protocol / Internet Protocol	Internet data transmission protocol

Phân biệt các độ lớn của các loại mạng:

Abbreviation	Full Name	Phạm vi	Typical Use Cases
LAN	Local Area Network	Nhỏ – trong một phòng / tòa nhà	Internal office network, classrooms, home networks
WLAN	Wireless Local Area Network	Giống LAN nhưng không dây (Wi-Fi)	Home Wi-Fi, cafés, wireless office setups
MAN	Metropolitan Area Network	Trung bình – một thành phố	Connecting offices across a city

WAN Wide Area Network	Rất rộng – liên tỉnh, liên quốc gia		Global enterprise networks, bank systems
Internet Global Network	Lớn nhất – toàn cầu		Accessing websites, email, social media, video streaming
Intranet Private Network	Nội bộ – bên trong một tổ chức		Internal company systems, document sharing
Extranet Extended Internal Network	Mở rộng Intranet – có kiểm soát		Secure partner/vendor access to company internal resources

#2 cái cuối là intranet và extranet không có phân biệt độ lớn

Vocabulary

The **router** lets a network connect to the Internet.

A LAN is a type of **network.**

Businesses communicate on the **Internet.**

We do not use cables for our network. We have a **WLAN.**

Routers need a correct **configuration** to work properly

Businesses with a LAN use **CAT-5** cable to connect computers.

It is sometimes necessary to **cycle** a network to make it work.

The **DHCP** provides different IP addresses for computers.

Chương 6-The User Interface

Từ	Full từ	Từ nhận diện
GUI Graphical User Interface	Graphic interface	(icons, windows, menus)
CLI Command Line Interface	Command input via keyboard	
UX User Experience	User experience	
UI User Interface	User interaction interface	
OS Operating System	System controller	

Vocabulary:

Use the **pointer** to choose files and commands

Small picture or symbols in software are **icons**

The is **operating system** the program that makes a computer word

When something is on the **desktop** you see it onscreen

An OS that uses small pictrure to repersent file is a **GUI**

flou can minimize file and programs in different **windows**

Chương 7-Word Processing

Abbreviation / Term	Full Form	Function / Role Keyword
WYSIWYG	What You See Is What You Get	Visual editing interface (Word, Google Docs)
RTF	Rich Text Format	Rich text formatting (bold, italic, colors)
DOCX	Document (Microsoft Word format)	Microsoft Word file format
PDF	Portable Document Format	Fixed-layout document, view-only

Vocabulary

Use a **macro** to type out phrases that you use a lot.

My file won't open in other programs due to a **compatibility** problem.

Instead of writing by hand, use a **word processing program**.

This file is in a **format** that doesn't open in TypeWrite.

Save each **document** with a different name so you have a copy of each.

The letter's **font** is wrong. The heading is in the wrong place

Increase the **font** size to make the file easier to read.

Most word processors offer **spell check** to help you find mistakes

Chương fi-Email

The _____ is a brief description of what the email is about. **Subject**

Abbreviation / Term	Full Form	Function / Role Keyword
Email	Electronic Mail	Digital message system
SMTP	Simple Mail Transfer Protocol	Sending emails
POP3	Post Office Protocol version 3	Downloading emails
IMAP	Internet Message Access	Email syncing across devices

	Protocol	
ISP Internet	Service Provider	Internet provider
CC Carbon Copy		Send visible copy
BCC Blind Carbon Copy		Send hidden copy
URL Uniform	Resource Locator	Embedded web link

Vocabulary:

A computer that sends email _ email clident

A program that saves, sends and receives email _ SMTP server

Words/letters used to identify companies and addresses _ domain name

Letters/numbers that let a person access something _ password

A file that is sent with an email _ attachment

Chương 9-Web Browsing

Vocabulary:

Hyperlink a word/image that you can click on to navigate

Website a collection of related Internet pages

HTML a computer language used to make web sites

web host a company that provides space for a website

web browser a program that lets you look at web pages

BOOK 2

Unit 1-Computer Memory

Vocabulary:

Where data that doesn’t fit in the RAM is stored_ **virtual memory**

The main circuit board in a computer _ **motherboard**

A unit of measurement that measures the speed of a CPU _ **MHz**

A data storage method that stores data on the hard drive _ **paging file**

The smallest unit of computer data _ **bit**

An memory module that sends up to 32 bits of data to a CPU _ **SIMM(Single Inline Memory Module)**

A memory module that sends up to 64 bits of data to a CPU _ **DIMM(Dual Inline Memory Module)**

Unit 2-Programming Languages

Vocabulary

compiler – language that computers can proces

Java – An object-oriented programming language is used often

assembler – A programming language that translates between program

BASIC – An early programming language that is not widely used anymore

C# – A competitor to Java

C++ – A programming language that adds capabilities to C

C – Used to program Linux

CPU – Central Processing Unit

compiler – A computer part that changes source code into machine code

Từ	Full từ	Nhận diện
IDE	Integrated Development Environment	programming tool, editor, compiler
OOP	Object-Oriented Programming	class, object, method, inheritance

Unit 3-ISPs and Internet Access

T-1– lines that allow many users to connect to the Internet

Wi-Fi– an Internet connection using wireless signals

ISP– a device that sends information over phone lines

ireless router– an Internet connection using regular phone lines

broadband– a system allowing the transfer of large amounts of data

DSL	Digital Subscriber Line	high-speed internet via telephone line

Unit 4-Storage Devices
Unit 5-Peripherals
Unit 6-MP3 Players
Unit 7-Cell Phones
Unit fi-Communications
Unit 9-Entertainment and Social Networking
Unit 10-GPS Systems
Unit 11-Banking
Unit 12-Education and Research.
Unit 13-Robotics
Unit 14-Electronic Publishing
Unit 15-Artificial Intelligence